motor-to-sensor FRFs: z = 0.5 mm

Closed-loop sweep; static geometric maps on logical axes. Status: converged.

Independently unwrapped; common 50 Hz branch convention; no smoothing. Each curve uses its own interpolated 50 Hz phase in [-270,90) deg.

Check-segment FRF

SRIVC model

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)